

<https://doi.org/10.1038/s41746-025-01443-2>

Preventing unrestricted and unmonitored AI experimentation in healthcare through transparency and accountability



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The integration of large language models (LLMs) into electronic health records offers potential benefits but raises significant ethical, legal, and operational concerns, including unconsented data use, lack of governance, and AI-related malpractice accountability. Sycophancy, feedback loop bias, and data reuse risk amplifying errors without proper oversight. To safeguard patients, especially the vulnerable, clinicians must advocate for patient-centered education, ethical practices, and robust oversight to prevent harm.

The addition of large language models (LLM) into healthcare operations, including electronic health record (EHR) systems represents a transformative level of adoption and integration, changing how enterprise systems interact with algorithms; permitting the largely unconsented collection and utilization of previously unmined private data originating from the care of patients^{1,2}. LLMs are a form of generative artificial intelligence (AI) capable of creating new content. Like many AI models, LLMs are “black boxes” due to their complex architecture, and are not easily interpretable³. Their opacity is exacerbated by the industry’s practice of maintaining proprietary control over these models, limiting access to the underlying algorithms and training data⁴. As a result, a lack of transparency can arise from both a model’s black box nature, and developer practices that limit a comprehensive understanding and verification of their potential risks and performance^{3–5}.

Their integration into EHR systems is characterized by broad scale, diverse applications, and an ongoing lack of oversight typical of software as a medical device and secondary data use, signaling an emerging and potentially transformative shift in care practice^{2,6}. However, the propensity of generative AI to display emergent behaviors and hallucinations, combined with the lack of comprehensive governance frameworks, let alone patient outcomes-centered validation systems, raises substantial concerns about the broad integration of LLMs into digital healthcare and the potential risks they pose to individuals, communities, and the broader public.

Lack of transparency and oversight in EHR AI integration

Without clear insight into the integration of AI models within EHR systems, distinguishing between AI applications for clinical care, quality programs (QPs), health system operations, or product enhancement becomes

challenging. This ambiguity hinders end-users and stakeholders from understanding the specific purposes and uses of AI in healthcare settings. While QPs involving human subjects are typically overseen by Institutional Review Boards (IRBs) as required by the Common Rule (45 CFR 46)⁷, clinical QPs usually adhere to accreditation standards set by organizations like the Joint Commission. However, the integration of AI and large language models (LLMs) into hospital EHR systems currently lacks a defined oversight framework (Table 1)². Additionally, there is no established ethical or legal requirement in the United States for obtaining informed consent for the use of medical AI in clinical practice. Existing ethical regulations governing research do not apply to the routine deployment of AI to improve operational efficiency or to reduce the cognitive load of healthcare providers which are not considered health products despite risk of harm to patient outcomes, raising significant ethical concerns^{8,9}.

Emerging risks and bias in LLM outputs

Although technologies are often intended to enhance workflows and throughputs and reduce clinician burnout, emerging evidence suggests that LLM use can lead to unintended harm. For example, outputs from LLMs, such as OpenAI’s ChatGPT, which uses GPT-4 (Generative Pre-trained Transformer 4)¹⁰—one of the most advanced text-based generative AI models¹¹—can be biased by patient demographics¹², leading to stereotypical presentations and potentially impacting the quality and equity of care¹³. For instance, assessments and plans generated by ChatGPT have been linked to recommendations for more expensive procedures, exacerbating healthcare disparities and costs¹². Additionally, clinical documentation produced by LLMs influences clinician decision-making through anchoring and automation biases^{14,15}. Increased awareness of these biases enables healthcare

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Table 1 | Stakeholder roles and responsibilities in LLM governance for healthcare

Stakeholder	Current Role	Actions Needed	Potential Conflict of Interest	Challenges	Accountability/Impact	Stakeholder Interests	Ethical Responsibilities	Potential Risks	Collaborative Opportunities
Healthcare Providers	Use AI in patient care and decision-making	Lead ethical AI use; advocate for transparency	Bias if AI favors efficiency over care quality	Limited AI literacy; over-reliance on AI	Direct impact on patient care and liability	Ensuring care quality, reducing liability	Ensure informed consent; prevent harm	Misdiagnosis, treatment errors	Work with developers on accuracy and ethics
AI Developers/Vendors	Design and deploy AI models	Increase transparency; follow ethical guidelines	Profit motive may hinder transparency	Balancing IP with need for transparency	Control over model reliability and trust	Adoption, profitability, competitive edge	Minimize bias; clarify AI decisions	Trust erosion, harmful decisions	Collaborate on ethical guidelines and validation
Institutional Review Boards (IRBs)	Oversee research ethics	Include AI's societal impacts; expand review scope	Pressure to approve rapidly; overlook societal impacts	Limited AI expertise; long-term impact assessment	Ethical oversight affects public trust	Ethical research, public trust	Protect subjects; assess societal impacts	Ethical breaches, public mistrust	Partner with ESR boards and AI experts
Hospitals/Institutions	Integrate AI into EHRs and operations	Develop AI governance; ensure transparency	Efficiency vs. care quality; financial ties to AI	Balance efficiency with patient care	Ethical AI use, patient outcomes, compliance	Efficiency, cost reduction, patient outcomes	Prioritize patient welfare; protect privacy	Compromised care quality, privacy issues	Collaborate on ethical deployment practices
Government Agencies	Regulate medical devices and AI	Establish AI regulations; enforce accountability	Balancing innovation with safety	Keeping pace with AI advancements	Public safety, compliance, fostering innovation	Safety, innovation, regulatory credibility	Ensure safety, transparency, prevent misuse	Unsafe AI, loss of public trust	Global collaboration on AI standards
Insurance Companies	Use AI for claims and cost-cutting	Ensure ethical AI use; disclose AI processes	Reducing costs at the expense of care	Avoid unfair claims denial; maintain profitability	Affects access to care and financial outcomes	Cost reduction, profit maximization	Fair claims processing; avoid care compromise	Denied treatments, customer mistrust	Partner on ethical AI claims processing
Patients/Caregivers	Often unaware AI participants in care	Advocate for transparency and consent	Limited AI awareness, potential exploitation	Understanding AI's role in care	Health outcomes, trust in healthcare	Health, privacy, equitable care	Demand transparency, ensure consent	Exploitation, privacy violations	Collaborate on rights-respecting AI use
Ethics & Society Review Boards	Assess societal impacts of AI	Establish ESR boards; ensure thorough evaluation	Underrepresentation risks, biased assessments	Comprehensive representation of societal impacts	Shape societal acceptance, ethical AI integration	Ethical AI integration, societal trust	Assess societal impacts; ensure inclusivity	Biased AI applications harming communities	Collaborate on societal impact assessments

Summarizes the roles, responsibilities, and challenges for stakeholders involved in the governance of Artificial Intelligence (AI) and Large Language Models (LLMs) in healthcare. Stakeholder: Key groups involved in AI governance, including healthcare providers, AI developers, IRBs, hospitals, government agencies, insurance companies, patients/caregivers, and ethics boards. Current Role: The existing responsibilities of each stakeholder in AI/LLM integration in healthcare. Actions Needed: Specific steps each stakeholder should take to promote ethical and transparent AI use. Potential Conflict of Interest: Possible conflicts, such as profit motives or balancing efficiency with care quality. Challenges: Obstacles faced, including limited AI literacy, balancing intellectual property, and staying current with AI advancements. Accountability/Impact: The influence of each stakeholder's actions on patient care, public trust, and AI success in healthcare. Stakeholder Interests: The primary concerns of each group, like ensuring care quality, minimizing bias, and fostering innovation. Ethical Responsibilities: The duties of each stakeholder, such as ensuring informed consent, preventing harm, and assessing societal impacts. Potential Risks: Risks associated with improper AI governance, such as misdiagnosis, compromised care, privacy violations, and erosion of trust. Collaborative Opportunities: Areas where stakeholders can collaborate to improve the ethical deployment of AI in healthcare.

providers to critically evaluate LLM outputs, mitigating potential harms and promoting more equitable patient care¹⁶.

Evolution and oversight of AI in healthcare

Previously, the integration of algorithms into health information systems focused on developing AI models to address specific clinical questions and workflow inefficiencies, such as Clinical Decision Support Systems (CDSS) for evidence-based recommendations, predictive analytics for forecasting patient outcomes, and automating complex coding and billing processes. These AI tools, built from limited subsets of clinical data extracted from medical records, were typically accessible only to healthcare professionals and billing specialists^{17,18}. Research involving conventional AI that falls under the Food and Drug Administration (FDA) definition of Software as a Medical Device is subject to oversight, primarily through FDA and IRB supervision, as the device undergoes testing and validation¹⁹. Commercial generative AI systems, such as those that use large language models (LLMs), were not initially designed as medical devices and therefore typically lack regulatory oversight throughout their development—from inception to validation. Classified as Software of Unknown Provenance (SOUP), these systems often do not disclose critical details about their data sources, development methodologies, or validation processes, making it challenging to assess and manage biases that may emerge at various stages of the model lifecycle^{8,20,21}. While AI systems not designed for medical purposes are not inherently subject to FDA or IRB regulation, concerns arise when these tools are repurposed for patient-facing or clinical decision-making tasks. Such applications may fall under FDA oversight if the AI is used for diagnosis, treatment, or disease prevention²⁰. Although the FDA applies discretionary enforcement policies for lower-risk software, where efficacy evidence may be minimal, the deployment of AI in high-risk contexts—such as interpreting medical data or guiding treatment—without thorough clinical evaluation introduces significant potential risks¹⁹. Ensuring rigorous, lifecycle-based oversight for AI used in clinical applications is essential to safeguarding patient safety.

Transparency and challenges in LLM development

LLMs can be trained on a variety of datasets, extending beyond web-scraped data to include proprietary datasets such as medical records, as demonstrated by models like GatorTron and ClinicalT5²². Additionally, LLMs are just one type of generative AI; other models, such as vision-language models (VLMs) and generative adversarial networks (GANs), can also be trained on diverse multi-modal datasets^{22,23}. However, there is often a lack of adequate transparency regarding the architecture, training methods, and datasets used to train many of the most popular commercial LLMs. This makes it more difficult to understand and anticipate risks compared to open-source models that provide more information about their model weights, training data, and methodologies, which may enable better understanding and verification of their capabilities and limitations^{24–26}.

Expanding applications and ethical challenges of LLMs in healthcare

While developers have provided high-level information about structure and functionality, critical elements such as data cards, model cards, and bias cards—proposed by AI researchers—are not always shared alongside a model. Although often not specifically trained for medicine, LLMs are now being utilized for a broader range of tasks, including medical language translation, patient communication, privacy protection, and workflow enhancement. They are also being integrated into CDSS and automating processes like discharge planning, coding, and billing^{6,27,28}. Despite their potential, issues such as the propagation of race-based medicine by LLMs have not yet been addressed in EHR implementations²⁷. However, several EHR vendors are moving forward with integrating generative AI across their platforms²⁹. For example, Epic announced plans to roll out more than 100 generative AI projects such as auto-adverse drug reaction tagging, patient-friendly report summaries, a hospital billing coding assistant and an “explain my bill” AI agent at the

2024 HIMSS conference³⁰. MEDITECH is collaborating with Google to leverage an LLM for drafting documentation notes that providers can edit before EHR incorporation. Similarly, Oracle Health’s AI “Clinical Digital Assistant” uses digital information and patient-provider conversations to draft notes and propose follow-up actions. AI is expanding EHR data sources by incorporating new types of data acquired from treatment rooms equipped with ambient monitoring systems. For example, technologies like the OR Black Box can capture detailed point-of-care data, including video, audio, and patient physiology, which can be utilized by LLMs for enhanced healthcare analysis^{31,32}.

Ethical implications and the need for transparency and accountability

In research, informed consent is typically obtained from both patients and providers. However, the influence of power dynamics between patients and investigators is often not fully addressed within the consent process. However, in clinical QP contexts, consent requirements can vary. Some institutions implement AI technology as part of routine QP activities without obtaining explicit consent for each use, instead relying on broader policies and general consent from patients for treatment and quality assurance⁹. Currently, there is no universally agreed-upon ethical or legal requirement to disclose the use of medical AI for informed consent in the United States, the European Union, or South Korea^{8,9}. The approach to consent varies by hospital policies, legal requirements, and ethical guidelines³³. Within this structure, patients and caregivers may become inadequately informed participants in the development of AI products, without ethical oversight from IRBs or Data and Safety Monitoring Boards³⁴. AI medical applications and tools present significant potential benefits as well as risks that extend beyond operational and administrative efficiencies. The widespread integration of these technologies into healthcare systems demands rigorous transparency and detailed disclosures to patients and clinicians regarding the data being collected, its intended use, and how it will be managed and shared.

Ethical concerns and conflicts of interest in LLM utilization for insurance claims

A recent class action lawsuit was filed against Humana for allegedly using the AI model NHPredict to wrongfully deny necessary care to Medicare Advantage patients³⁵. Another lawsuit claims that United Healthcare used the same model to override clinicians’ medical determinations, leading to the rejection of valid claims³⁶. The use of LLMs by insurance companies for claim processing and fraud detection raises significant ethical concerns³⁷. Currently, several LLMs, including versions of the ChatGPT family, are employed by both EHR systems and insurance companies^{38–40}, creating a conflict of interest when similar datasets are used by entities with opposing objectives^{21,41}. For example, insurance companies use LLMs to minimize costs by detecting fraud and reducing claim payouts¹⁸, while medical institutions use them to optimize billing and maximize reimbursements. This dual use of LLMs can lead to biased outcomes, where patient care decisions are influenced more by financial considerations than medical necessity. Consequently, legitimate claims may be unfairly denied, underscoring the need for careful oversight of LLM use in these contexts.

The shared use of EHRs for conflicting purposes exacerbates these ethical concerns. It’s crucial to ensure that patient data is used to improve care, not merely to reduce costs. The differing applications of AI by hospitals and insurers could result in inconsistent care delivery—hospitals might leverage AI to justify necessary treatments, while insurers use it to minimize expenditures.

To maintain trust and accountability, there must be transparency in how data and models are used by different entities. Regulatory bodies should oversee the deployment of LLMs to ensure fair and unbiased application. Establishing ethical frameworks and oversight committees for AI in healthcare is essential to address conflicts arising from the dual use of data and models.

Ethical considerations and accountability in AI-driven patient management

AI-driven predictive models, such as those identifying patients at risk of “no show” visits, illustrate the broader application of algorithmic decision-making in healthcare. While these models typically rely on structured data, the principles of transparency and ethical oversight apply universally across AI systems, including LLMs. The adaptation of AI to influence appointment scheduling raises important concerns about algorithmic bias, the fairness of overbooking practices, and the potential for exacerbating disparities in care. As with LLMs, there is currently limited visibility for providers and patients into the decision-making processes of these models, highlighting the need for robust oversight and ethical frameworks to guide AI-mediated care decisions.

To address these concerns, it is crucial that both providers and patients have access to information about when algorithms are used in decision-making processes, as well as information on how those algorithms were developed and validated. Additionally, there must be accountability for the patient outcomes of these AI-driven decisions, with continuous monitoring and validation to ensure they do not perpetuate inequities in healthcare delivery.

Currently, at many institutions, patients and clinicians cannot opt out of AI-driven quality improvement (QI) processes, research, or clinical care. AI vendors have not provided clear information on how they will establish safeguards to protect patient privacy and ensure that clinicians are fully aware of what data is being recorded about their actions, how this data is accessed, and how it may be linked to patient care or potential malpractice and legal claims. Additionally, they have not clarified how they will prevent the misuse of EHR data that could lead to unfair or inaccurate restrictions on services.

Ensuring security, accountability, and ethical oversight in AI integration

Hospitals are legally mandated to secure identifiable patient data, especially when shared with third-party AI vendors⁴². However, integrating LLMs into healthcare systems introduces risks such as prompt injection attacks, where malicious actors manipulate the AI's behavior, data memorization, which risks retention of sensitive information, and arbitrary code execution. While ongoing research addresses these risks through enhanced red-teaming practices and guardrail standardization^{43,44}, these solutions are still in early stages and are not yet fully operational or widely implemented⁴⁵.

A significant gap in the current landscape is the lack of rigorous review of FDA-approved medical algorithms post-deployment, which should engage clinicians, ethicists, patients, and caregivers. Fewer than 30% of AI tools have had their performance and effectiveness thoroughly scrutinized^{46,47}. This oversight gap can lead to security and privacy vulnerabilities, making a more expansive and inclusive review essential to ensure these algorithms are safe and effective before widespread use. A multi-stakeholder assessment is crucial to validating these algorithms both before and after deployment^{46,47}.

Currently, little responsibility is placed on AI developers to predict, mitigate, and report the risks their algorithms may cause. There is no established accountability chain for algorithmic malpractice, leaving clinicians exposed to liability, especially when AI systems are used to identify and track medical errors. For instance, if a radiologist flagged by an AI-enabled QP review later contributes to a wrong-side procedure due to inaccurate AI analysis, the implications for liability are unclear. Healthcare systems are not yet prepared to manage the complex issues arising from widespread AI use, particularly when the AI itself may be flawed.

Moreover, providers' actions as they interface with AI tools are increasingly monitored and analyzed without fully informed consent, as surveillance processes are embedded in imaging platforms, EHR systems, and QP initiatives—all of which lack adequate safeguards⁴⁸. To address these challenges and protect both clinician and patient data, robust access controls, audit trails⁴², transparency standards, and encryption measures are necessary⁴⁹. Transparency in monitoring practices and the establishment of disclosure management systems are crucial, allowing providers to opt-in or

out of monitoring where possible^{42,48}. Anonymization and regular security assessments in line with regulations like the Health Insurance Portability and Accountability Act are also essential^{42,49}.

Ethical review boards should oversee clinical and research AI use, ensuring adherence to data privacy and security best practices^{42,50}. Unlike many international ethics bodies, U.S. IRBs exclude societal harms from their review criteria, which may prevent long-term effects from hindering beneficial research. However, this approach risks overlooking broader implications. Federal regulations explicitly state that IRBs should not consider the long-range effects of research on public policy, which may inadequately address the ethical and societal risks arising from AI's broader impacts⁵¹.

Clinician leadership in addressing algorithmic bias in healthcare

While entities such as the World Health Organization and the Biden administration have developed criteria for safe, fair, and transparent AI, comprehensive AI governance systems that function enterprise-wide are still lacking in many academic institutions, companies, and startups. Although providers currently bear much of the liability for AI malpractice, patients face risks to their health, privacy, and dignity. As medical AI becomes more prevalent, both patients and clinicians must educate themselves about its development and associated risks⁴⁸. AI vendors must also ensure transparency, enabling end-users to understand and critically evaluate the decisions made by these models⁵⁰.

In 2022, the Department of Health and Human Services proposed extending the Affordable Care Act to include protections against biases in clinical algorithms. While this initiative aims to combat disparities, it has raised concerns about placing an unfair burden on healthcare professionals, requiring them to assess the fairness and legality of algorithms—tasks that often exceed their expertise^{52,53}. This issue is compounded by the lack of shared accountability for AI malpractice. Currently, the burden of liability falls disproportionately on practitioners, with insufficient responsibility placed on vendors and institutions. This imbalance exacerbates the challenges of navigating the proprietary and opaque nature of many AI models used in healthcare, which often lack explainability and transparency, making it difficult to evaluate and address potential biases. Research by Gichoya et al. has shown that machine learning systems can predict racial identity from de-identified medical images, raising ethical concerns, particularly given the history of racial bias in U.S. healthcare. Similar issues with gender inference further underscore the need for vigilant oversight²⁸.

Clinicians, as the primary stewards of patient care, must lead the effort to ensure ethical AI integration. The American Medical Association advocates for shared responsibility among AI developers, institutions, and users, but physicians and other clinicians must be at the forefront, championing transparency, accountability, and robust ethical guidelines. Physician involvement is essential to ensure that AI enhances, rather than undermines, patient care and that liability for AI-driven decisions is appropriately distributed^{53–56}.

Conclusion: the imperative for AI governance, education, and clinician leadership

The integration of LLMs into EHRs, while enhancing operational efficiency, presents significant challenges that require careful evaluation of both benefits and risks⁵⁷. The current absence of robust AI governance in healthcare places liability on providers and exposes patients to potential harm. To mitigate these risks, healthcare institutions should demand clear protocols and comprehensive disclosures from AI developers and EHR vendors, detailing when, where, how, and why AI is used, along with the specific features involved.

Clinician leadership is vital in navigating these challenges. As advocates for patient welfare, clinicians must actively guide the ethical integration of AI into clinical practice by championing the development of local AI governance frameworks that ensure AI systems are deployed safely and equitably.

Developing such governance requires collaboration among vendors, medical institutions, IRBs, government agencies, patient advocacy groups, and licensing organizations to foster large-scale educational initiatives that enhance AI literacy among the public and clinicians. Institutions could establish Ethics and Society Review boards, like Stanford Health's, to examine the societal impacts of AI projects⁵⁸. Additionally, patients and providers must be educated on safely interacting with AI, with an option to opt out of AI use when transparency around algorithm development and validation is lacking. AI vendors should be required to provide training on AI bias and algorithmic robustness to enable users to critically evaluate these technologies.

Finally, healthcare providers must educate themselves on the ethical deployment of AI to protect patients and uphold the integrity of medical practice amid rapid technological advancements. Clinician leadership, coupled with robust governance and education, is crucial to ensuring AI enhances rather than undermines the quality of patient care.

Received: 3 May 2024; Accepted: 8 January 2025;

Published online: 18 January 2025

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Acknowledgements

D.S.B. is funded by the National Institute of Health through NIH-USA U54CA274516-01A1 and NIH-USA R01CA294033-01. L.A.C. is funded by the National Institute of Health through NIBIB R01 EB017205.

Author contributions

D.S.C. conceptualized the perspective piece and led the writing process. D.S.B. and L.A.C. contributed to the development of ideas and participated in drafting and revising the manuscript. D.S.C. provided critical revisions and refined the final draft. All authors (D.S.C., D.S.B., and L.A.C.) reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

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